

CO2 - AT1 - Database Design and Connectivity

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COURSE NAME:DSA0503 - Query Processing for Data
Science

Question 1: Online Course Registration System

Aim:

To create Student and Course_Registration tables, insert sample records, and display the names of students along with the courses they have registered for using an SQL JOIN operation.

Algorithm:

1. Import the SQLite and Pandas libraries.
2. Create a SQLite database connection.
3. Create the Student table.
4. Create the Course_Registration table with a foreign key.
5. Insert sample student records.
6. Insert course registration records.
7. Join the Student and Course_Registration tables.
8. Display the student names and registered courses.
9. Close the database connection.

CODE:

```
import sqlite3
```

```
import pandas as pd
```

```
conn = sqlite3.connect("CourseDB.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Student(
```

```
    Student_ID INTEGER PRIMARY KEY,
```

```
    Student_Name TEXT NOT NULL,
```

```
    Department TEXT NOT NULL
```

```
)
```

```
""")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Course_Registration(
```

```
    Registration_ID INTEGER PRIMARY KEY,
```

```
    Student_ID INTEGER,
```

```
    Course_Name TEXT NOT NULL,
```

```
    Semester INTEGER,
```

```
    FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID)
```

```
)
```

```
""")
```

```
cursor.execute("DELETE FROM Course_Registration")
```

```
cursor.execute("DELETE FROM Student")
```

```
students = [
```

```
(101,"Arun","CSE"),  
(102,"Priya","AI&DS"),  
(103,"Rahul","ECE"),  
(104,"Sneha","IT")  
]
```

```
cursor.executemany("INSERT INTO Student VALUES(?,?,?)",students)
```

```
courses = [  
    (1,101,"Python Programming",3),  
    (2,102,"Database Systems",4),  
    (3,103,"Computer Networks",5),  
    (4,104,"Machine Learning",6)  
]
```

```
cursor.executemany("INSERT INTO Course_Registration  
VALUES(?,?,?,?)",courses)
```

```
conn.commit()
```

```
query = """
```

```
SELECT Student.Student_Name,  
       Student.Department,  
       Course_Registration.Course_Name,
```

```

        Course_Registration.Semester
FROM Student
INNER JOIN Course_Registration
ON Student.Student_ID = Course_Registration.Student_ID
"""

df = pd.read_sql_query(query,conn)

print("Student Course Registration Details\n")

print(df)

conn.close()

```

OUTPUT:

```

... Student Course Registration Details

```

	Student_Name	Department	Course_Name	Semester
0	Arun	CSE	Python Programming	3
1	Priya	AI&DS	Database Systems	4
2	Rahul	ECE	Computer Networks	5
3	Sneha	IT	Machine Learning	6

Question 2: Airline Reservation Management System

Aim:

To create Passenger and Booking tables, insert booking records, and display passenger names along with their booked flight details using an SQL JOIN operation.

Algorithm:

1. Import the SQLite and Pandas libraries.
2. Create a SQLite database connection.
3. Create the Passenger table.
4. Create the Booking table with a foreign key.
5. Insert passenger records.
6. Insert booking records.
7. Join the Passenger and Booking tables.
8. Display passenger names with flight details.
9. Close the database connection.

CODE:

```
import sqlite3
```

```
import pandas as pd
```

```
conn = sqlite3.connect("AirlineDB.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Passenger(
```

```
    Passenger_ID INTEGER PRIMARY KEY,
```

```
    Passenger_Name TEXT NOT NULL,
```

```
    Gender TEXT
```

```
)
```

```
""")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Booking(  
    Booking_ID INTEGER PRIMARY KEY,  
    Passenger_ID INTEGER,  
    Flight_No TEXT NOT NULL,  
    Source TEXT,  
    Destination TEXT,  
    FOREIGN KEY(Passenger_ID) REFERENCES  
    Passenger(Passenger_ID)  
)  
""")
```

```
cursor.execute("DELETE FROM Booking")
```

```
cursor.execute("DELETE FROM Passenger")
```

```
passengers = [  
    (101,"Arun","Male"),  
    (102,"Priya","Female"),  
    (103,"Rahul","Male"),  
    (104,"Sneha","Female")  
]
```

```
cursor.executemany("INSERT INTO Passenger  
VALUES(?,?,?)",passengers)
```

```
bookings = [  
    (1,101,"AI101","Chennai","Delhi"),  
    (2,102,"AI202","Bangalore","Mumbai"),  
    (3,103,"AI303","Hyderabad","Kolkata"),  
    (4,104,"AI404","Chennai","Goa")  
]
```

```
cursor.executemany("INSERT INTO Booking  
VALUES(?,?,?,?)",bookings)
```

```
conn.commit()
```

```
query = ""
```

```
SELECT Passenger.Passenger_Name,  
    Passenger.Gender,  
    Booking.Flight_No,  
    Booking.Source,  
    Booking.Destination  
FROM Passenger  
INNER JOIN Booking  
ON Passenger.Passenger_ID = Booking.Passenger_ID  
""
```

```
df = pd.read_sql_query(query,conn)
```

```
print("Passenger Booking Details\n")
```

```
print(df)
```

```
conn.close()
```

OUTPUT:

```
.. Passenger Booking Details
```

	Passenger_Name	Gender	Flight_No	Source	Destination
0	Arun	Male	AI101	Chennai	Delhi
1	Priya	Female	AI202	Bangalore	Mumbai
2	Rahul	Male	AI303	Hyderabad	Kolkata
3	Sneha	Female	AI404	Chennai	Goa